REFINERY CENTRIFUGAL PUMP OPERATIONS MANUAL

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Asset Class: Centrifugal Pump

Purpose: Refinery Fluid Transfer Service

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1. Purpose

This manual defines the operating philosophy, process function, telemetry interpretation, alarm logic, operating limits, and reliability considerations for refinery centrifugal pumps.

The document serves as the primary operational reference for:

- Operators
- Maintenance Engineers
- Reliability Engineers
- Process Engineers
- Industrial AI Copilot

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2. Equipment Overview

A centrifugal pump converts rotational mechanical energy into fluid flow and pressure.

The pump receives fluid through the suction nozzle and accelerates the fluid using a rotating impeller.

The kinetic energy generated by the impeller is converted into pressure energy within the casing.

The pump is responsible for:

- Crude transfer
- Product transfer
- Cooling water circulation
- Process feed circulation
- Utility fluid transport

Pump reliability directly impacts refinery throughput, equipment availability, and product quality.

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3. Major Components

3.1 Impeller

Function:

Transfers energy to the process fluid.

Failure Impact:

- Reduced flow rate
- Reduced discharge pressure
- Reduced efficiency

3.2 Shaft

Function:

Transfers torque from the motor to the impeller.

Failure Impact:

- Vibration increase
- Seal damage
- Bearing overload

3.3 Bearings

Function:

Support rotating shaft.

Failure Impact:

- Elevated vibration
- Increased temperature
- Increased motor current
- Catastrophic failure risk

3.4 Mechanical Seal

Function:

Prevent process fluid leakage.

Failure Impact:

- Product loss
- Safety hazard
- Environmental release

3.5 Electric Motor

Function:

Provides rotational energy.

Failure Impact:

- Increased current draw
- Temperature rise
- Unexpected shutdown

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4. Telemetry Philosophy

The Industrial AI Copilot continuously monitors the following telemetry signals.

Each parameter represents a different aspect of equipment health.

Multiple parameter deviations occurring simultaneously typically indicate a developing failure mode.

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5. Telemetry Parameter Definitions

5.1 Bearing Temperature

Telemetry Field:

bearing_temp

Description:

Measures thermal condition of pump bearings.

Engineering Importance:

Bearings generate heat when friction increases.

Elevated temperature is often the earliest indication of bearing degradation.

Normal Range:

50°C – 75°C

Warning:

75°C – 85°C

Critical:

85°C

Trip Recommendation:

95°C

Common Causes of High Temperature:

- Lubrication deficiency
- Bearing wear
- Misalignment
- Excessive load

Operational Impact:

- Reduced bearing life
- Increased vibration
- Potential seizure

5.2 Motor Temperature

Telemetry Field:

motor_temp

Description:

Measures thermal loading of the electric motor.

Normal Range:

55°C – 85°C

Warning:

85°C – 95°C

Critical:

95°C

Common Causes:

- Overload
- Bearing drag
- Electrical imbalance
- Poor ventilation

Impact:

Motor insulation degradation.

5.3 Vibration

Telemetry Field:

vibration

Description:

Measures mechanical instability.

Normal Range:

0 – 4 mm/s

Warning:

4 – 7 mm/s

Critical:

7 mm/s

Failure Indicators:

- Bearing wear
- Misalignment
- Cavitation
- Rotor imbalance

Importance:

Vibration is among the most powerful predictors of future failure.

5.4 Flow Rate

Telemetry Field:

flow_rate

Description:

Fluid delivered by the pump.

Normal Range:

250 – 350 m^3/hr

Warning:

200 – 250 m^3/hr

Critical:

< 200 m^3/hr

Common Causes of Reduction:

- Cavitation
- Seal leakage
- Impeller damage
- Suction restriction

Business Impact:

Reduced production throughput.

5.5 Suction Pressure

Telemetry Field:

suction_pressure

Description:

Pressure at pump inlet.

Normal Range:

2 – 5 bar

Critical:

< 2 bar

Importance:

Low suction pressure is a primary contributor to cavitation.

5.6 Discharge Pressure

Telemetry Field:

discharge_pressure

Description:

Pressure generated by pump.

Normal Range:

8 – 15 bar

Critical:

< 8 bar

Possible Causes:

- Internal leakage
- Seal failure
- Impeller degradation

5.7 Motor Current

Telemetry Field:

motor_current

Description:

Electrical current consumed by motor.

Normal Range:

80 – 120 A

Warning:

120 – 140 A

Critical:

> 140 A

Importance:

Often rises before mechanical failures become visible.

5.8 Lubrication Level

Telemetry Field:

lubrication_level

Description:

Indicates availability of bearing lubricant.

Normal Range:

70 – 100%

Warning:

50 – 70%

Critical:

< 50%

Impact:

Low lubrication accelerates bearing degradation.

5.9 Seal Leakage Rate

Telemetry Field:

seal_leakage_rate

Description:

Measures fluid leakage through sealing system.

Normal Range:

0 – 0.1 L/hr

Warning:

0.1 – 0.5 L/hr

Critical:

0.5 L/hr

Safety Impact:

Potential environmental release.

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6. Telemetry Correlation Patterns

Pattern A

Bearing Temperature ↑

Vibration ↑

Motor Current ↑

Interpretation:

Likely Bearing Failure

Confidence:

High

Pattern B

Seal Leakage ↑

Flow Rate ↓

Discharge Pressure ↓

Interpretation:

Mechanical Seal Failure

Confidence:

High

Pattern C

Suction Pressure ↓

Vibration ↑

Flow Rate ↓

Interpretation:

Cavitation

Confidence:

High

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7. Alarm Philosophy

The objective of alarms is to provide sufficient time for intervention before production loss or equipment damage occurs.

Alarm Priority Levels:

Level 1:

Operator Awareness

Level 2:

Maintenance Planning

Level 3:

Immediate Intervention

Level 4:

Emergency Shutdown

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8. AI Copilot Interpretation Rules

Failure Probability > 80%

AND

RUL < 30 Days

Interpretation:

High Risk Asset

Recommended Action:

Create Maintenance Work Order

Failure Probability > 90%

AND

RUL < 15 Days

Interpretation:

Critical Asset

Recommended Action:

Immediate Reliability Review

Failure Probability > 95%

AND

RUL < 10 Days

Interpretation:

Potential Imminent Failure

Recommended Action:

Emergency Maintenance Planning

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9. Summary

Pump health is determined through the combined interpretation of:

- Bearing Temperature
- Motor Temperature
- Vibration

- Flow Rate
- Suction Pressure
- Discharge Pressure
- Motor Current
- Lubrication Level
- Seal Leakage Rate

No single parameter should be evaluated in isolation.

The highest confidence diagnoses are achieved by correlating multiple telemetry signals, failure probability predictions, failure mode classifications, and remaining useful life estimates.